Reading the prototype, screen by screen

What every tab shows, what each number means, why it is there at all, and what it does that a survey van, a crack-detection app or a complaint portal cannot. Every figure below came out of a real eighteen-photograph run.

A WALKTHROUGH FOR ANYONE SEEING IT FOR THE FIRST TIME

CONTENTS

§ First: what *is* the GPS number?

When a phone takes a photograph with location switched on, it does not just save the picture. In the same file — in a hidden section called **EXIF**, which every camera writes and almost nobody looks at — it records where the camera was standing at the instant the shutter fired.

Here is every number RAAHI pulled out of the *first* photograph in the demo round. Nothing here was typed in. It was already inside the file.

HIDDEN INSIDE DAY 1 OF THE ROUND	
GPSLatitude	`28.613969° N` How far north of the equator. Delhi sits at about 28.6.
GPSLongitude	`77.209064° E` How far east of Greenwich. Together, these two numbers name one point on Earth.
in degrees·min·sec	`28° 36' 50.29" N   77° 12' 32.63" E` The same fix written the way a survey drawing writes it, so an engineer recognises it.
GPSAltitude	`216.0 m` Height above sea level.
GPSImgDirection	`98.68° from true north` Which way the camera was **pointing** — nearly due east. This matters more than you would think.
GPSHPositioningError	`± 4.5 m` The phone's own honesty: how far out it thinks it might be.
DateTimeOriginal	`2026-08-21 06:52` When the shutter fired. This is what dates the reading — not when you uploaded it.
FocalLengthIn35mmFilm	`80 mm` How wide the lens was. This is what turns pixels into millimetres.

How precise is that, really?

Six decimal places of latitude is about **11 centimetres** on the ground. But the phone's *own* error bar says ± 4.5 m. So the number is written very precisely and known only roughly — and RAAHI never pretends otherwise. It shows the accuracy figure right next to the coordinates.

THIS IS THE PROBLEM THE WHOLE PROJECT TURNS ON

Two cracks four metres apart are different cracks. A GPS fix good to ± 5 m cannot tell them apart. So **GPS alone is not enough**, and any judge who knows GPS will say so within thirty seconds. RAAHI's answer to that is on the CAPTURE tab below.

And when the photo has no GPS?

A webcam frame has no metadata at all. So at the moment RAAHI takes a picture, it **writes the position in itself** — latitude, longitude, altitude, heading, the UTC satellite clock and the accuracy — into the file's own EXIF, exactly as a phone would. Afterwards you can copy that file anywhere, open it in any photo viewer, and the coordinates are there.

SAY THIS OUT LOUD

“The photograph carries its own evidence. Open it in any EXIF viewer and the coordinates are in the file — so the record and the pixels can never drift apart, which is what a municipality would need if a repair bill were ever disputed.”

---

00 Overview

What you see. A sunrise over a road, drawn entirely in code — no image file. Under it, two short lists: what this build does, and what it does not. At the bottom, a live count of where the round stands.

Why it is there. This page exists to set expectations before anyone touches anything. The second list is deliberate and it is the more important of the two.

> **THE LIMITS LIST IS YOUR ARMOUR, NOT YOUR WEAKNESS**
>
> Every prototype has holes. A panel that *tests a limit you already named* goes far better than one that *finds a limit you hid*. Naming them first also means the questions arrive on your terms — you have already answered them.

The four limits, and the honest answer to each

STATED LIMIT	WHAT YOU SAY WHEN ASKED
No trained classifier yet	The crack is found by image processing — thresholding, morphology, connected components. We measure growth; classification is the next module, and RDD2022 is the dataset for it.
Rainfall is a normal, not a forecast	A normal says what a September fortnight *usually* brings at that station. The app labels it `normal` and never calls it a forecast. Hand it a real IMD figure and it uses that instead.
Traffic uses class defaults	Until a municipality gives us counts. Every answer says whether the figure was counted or assumed.
Daylight, close range	Camera roughly overhead. Not heavy rain. This is a limitation of the optics, and we state it rather than discover it on stage.

01 Capture

This is where everything enters the system. Four things happen to every photograph, in order, in about half a second.

1 · The crack is found without being told where it is

The browser converts the photo to grey, sweeps a darkness threshold across it, cleans up the speckle a grainy road throws off, then groups the surviving dark pixels into connected shapes. It scores each shape on how *crack-like* it is: thin, long, near the middle, not running off the edge, clearly darker than the tarmac around it. The winner is outlined in red on screen.

If it picks the wrong thing, **"Wrong shape? Next candidate"** steps through the runners-up. That button is worth showing — it says the system knows it can be wrong.

2 · Pixels become millimetres

Nobody is asked to calibrate anything. The app works the scale out itself, from the best source it has, and always says which one it used:

THE SCALE LADDER — BEST SOURCE FIRST	
1 · site	a calibration measured at this exact spot
2 · global	a calibration from the training tab, rescaled if the photo is a different size
3 · the lens	focal length × shooting distance Pure optics. An 80 mm-equivalent lens at 1.2 m covers a known width of road, so one pixel is a known number of millimetres. No ruler needed.
4 · assumed	a 26 mm phone lens Only when the photo carries no lens data — a stripped file, a webcam frame. Every reading it produces says **assumed** on its face.

3 · The position is read — or written

The panel **WHERE THIS PHOTO WAS TAKEN** fills in with every number from the GPS section above, plus a green badge saying **FROM THE PHOTO** when the camera supplied it, or **FROM THIS DEVICE** when the browser did. There is a map link, so the claim can be checked against the world in one click.

4 · "Is this yesterday's crack?" — the hardest question in the project

GPS alone cannot answer it. So RAAHI votes with **three independent signals**:

SIGNAL	WHAT IT RULES OUT	TOLERANCE
Distance `haversine between two fixes`	A different street entirely	`15 m`
Heading `GPS image direction`	Standing in one place and photographing two different edges of the road	`45°`
Appearance `64-bit difference hash`	Everything else — and it works with no GPS at all	`12 of 64 bits`

A photo is the same spot when **position and heading agree**, or when the **scene alone is unmistakable**. And every single match is explained in plain words on screen. This is the sentence the app printed for day 2 of the real run:

POINT AT THIS ON SCREEN

"1.2 m from the recorded position; camera pointing within 4 deg of before; scene matches (1 of 64 bits differ)"

The difference hash deserves a word. It records whether each coarse patch of the picture is brighter than the patch to its right. That makes it **blind to exposure** — the thing that changes most between a 7 a.m. and a 9 a.m. pass — while staying sensitive to the layout of the scene. Across all eighteen demo passes, shot at deliberately different exposures, the fingerprint never differed by more than `2 bits of 64`.

> **EACH VISIT MAKES TOMORROW'S MATCH TIGHTER**
>
> After every pass the spot is re-centred on the *average* of its own fixes. More visits, smaller error in the recorded position, easier match next morning.

The shutter that presses itself

Press **Capture photo** and the camera watches its own preview once a second. When something crack-shaped holds still for **three frames running**, it fires by itself and stamps the GPS into the file at that instant.

Three frames, not one, because a shadow crossing the lens or a passing wheel is gone by the next frame. And this is the point of the whole deployment idea: **a driver on a collection round has both hands on the wheel.** Nobody is going to press anything.

Two controls worth demonstrating

- **Choose several photos at once.** Two, five, eighteen, a season. They upload in order with a line each. Order matters — photo two has to find photo one already in the record before it can be recognised as the same place.
- **"This is a different crack."** Two cracks a metre apart sit inside every tolerance the matcher has. The only thing that can separate them is the person holding the camera. Tick it and the next upload opens its own spot, leaving everything already recorded untouched.

02 Spots

Every place a photograph has been taken, with the position the app will match against tomorrow, and how many passes each has. Baseline and latest side by side.

This is the tab that shows the system is keeping a *register of assets*, not a pile of photos. Each row is a piece of road with a history.

At the bottom, **CLEAR EVERY RECORD** empties the round — spots, readings and stored photographs — so you can demonstrate twice in a row from nothing. Calibration and the seal threshold survive, because those describe the camera and the policy rather than the road.

03 Growth

Everything before this tab, other systems can also do. **This is the tab nobody else has**, and it is where you should spend your time.

The seal light

One verdict, in colour, above everything else. A ward engineer does not read a growth rate — they read *whether to send a crew this week*.

SEAL IT
Past the threshold, or crossing it within a fortnight. The date is printed on the right.

WATCH IT
Growing, with margin. Keep photographing; not a crew job yet.

HOLDING
Not growing across the passes recorded. Nothing to seal here.

NOT MEASURED TWICE
One photograph, or all on one day. No rate, so no colour.

> **GREY IS NEVER GREEN — AND THIS IS A DELIBERATE DESIGN DECISION**
>
> A spot photographed once is **grey**, not green. "We never went back" is not the same claim as "we checked and it is fine". Colouring it green would be the single most dishonest thing this app could do, and every road authority dataset in the country makes exactly that mistake by omission.

Baseline against latest

Day one and the newest pass, full size, side by side, with the difference in millimetres between them. In the real run: `110.1 mm → 134.3 mm`, a growth of `+24.2 mm`.

WHY GROWTH, AND NOT LENGTH

Absolute length carries a small constant bias — a thick stroke's outline adds its own width, every time. **In a subtraction a constant bias cancels exactly.** So growth survives a biased ruler where length does not. That is why growth is the number this app sells, and it is a genuinely good answer to "how do we trust your millimetres?"

The curve, and the numbers under it

Every pass plotted against the red dashed **seal threshold** line, with a faint fitted line through them. The threshold is a slider — set it to whatever the ward's policy says. Below the chart, six figures from the real eighteen-day run:

LIVE FIGURES · 18 PHOTOGRAPHS OVER 17 DAYS	
GROWTH	+24.2 mm Baseline to latest.
PER WEEK	+9.74 mm The rate a maintenance schedule can be built on.
LEAD TIME	28.3 days First sighting to predicted failure. The figure nobody in India publishes.
DAYS LEFT	11.3 days From today, at the measured rate.
FIT R^2	0.974 How well a straight line fits the readings. 1.00 is perfect. Below 0.6 the app prints a warning and calls the date rough — it does not hide a bad fit.
PASSES	18

THE SINGLE MOST IMPORTANT SENTENCE IN YOUR PITCH

“Lead time only exists if somebody photographs the same square metre of road more than once. That is why no Indian road authority reports it today — and it is the one output of this prototype that is not available anywhere else.”

---

04 Seal list

Two cracks growing at the same rate are not equally urgent. One is under a monsoon sky on a truck route; the other is on a dry residential lane. So the list is ordered by the formula from the deck:

1.392	×	2.17	×	1.00	=	77
MM / DAY · MEASURED		RAIN · 70 MM IN 14 DAYS		TRAFFIC · RESIDENTIAL LANE		PRIORITY

Read left to right. **Only the first term is measured** — it comes from photographs of that actual crack. The other two are stated multipliers, each with its own reference value, and the app prints the arithmetic so anyone can check it by hand.

WHERE EACH TERM COMES FROM	
`growth`	`1.392 mm/day` Least-squares fit over the 18 readings stored for this spot. Measured, not modelled.
`rain`	`1 + mm / 60, capped at 5` 70 mm expected over the next fortnight at New Delhi → factor 2.17. One in a dry fortnight, so a dry-season priority is growth × traffic and nothing invented.
`traffic`	`√(commercial vehicles ÷ 300)` A residential lane is exactly 1.0. An urban arterial is 2.58, a national highway 4.0. Commercial vehicles, because it is axle load — not vehicle count — that breaks a road.
`priority`	`0 – 100, logarithmic` 0.05 mm/day effective scores 0; 10 mm/day scores 100. Logarithmic because the product spans orders of magnitude — a straight line would pin everything worth looking at to 100 and the ranking would stop ranking.

The number that wins the argument

The list prints two crossing dates for the same crack:

WHAT THE MONSOON COSTS	
dry weather	18 September
with the rain	12 September
difference	6 days Sealing before the rain is not the same job as sealing after it. This is slide 2 of your deck, turned into a date.

BE FIRST TO SAY THE CONSTANTS ARE CHOSEN

60 mm a fortnight, the 0.5 traffic exponent, the score's floor and ceiling — these are **chosen, not fitted**. The app returns them under `assumptions` in its own output so nobody has to read the source to find out. They are the first thing to re-fit against a season of real readings, and saying so before you are asked is worth more than any answer afterwards.

A spot with only one photograph is **not scored at all**. It falls to the bottom in one block rather than being interleaved with real results — because a list that implied "considered and found safe" about a crack nobody re-visited would be worse than no list.

Report

The five boxes of the deck, on one screen, for a single crack. This is the tab to open when someone says *"show me the whole loop"*.

DETECT → TRACK → PREDICT → SCHEDULE → VERIFY	
1 · DETECT	18 photographs · 110.1 → 134.3 mm Every one taken, measured and stored on this machine. Nothing simulated.
2 · TRACK	18 of 18 carried a GPS fix The position in decimal and in degrees-minutes-seconds, the match radius, a map link.
3 · PREDICT	1.392 mm/day → 18 Sept The rate, the crossing date, and the 28.3-day lead time.
4 · SCHEDULE	priority 77 · URGENT The formula worked out term by term, and the seal-by date with the rain in it.
5 · VERIFY	ON TREND "The last pass gained 1.4 mm against an expected 1.4."

Being straight about VERIFY

The deck promises repair verification on tomorrow's collection round. This build has **no repair record**, so it reports the one thing the readings can honestly support: whether the last pass continued the trend or broke it.

A length that *stops growing* after a crew visited is what a seal looks like from here. A length that *jumps* is what a failed repair looks like. The app says which, and says when it cannot say — verification needs at least three measured passes.

Below the five boxes, **every pass day by day** is folded behind a summary line — day number, date, length, growth, position, and the sentence explaining why that photo was matched to this spot. Open it only if someone asks. Eighteen rows of detail is a gift to a hostile question in a five-minute demo.

At the top of this tab you can set the **road class** and a real vehicle count. Change it from residential lane to urban arterial and watch the traffic term move from 1.00 to 2.58 and the priority climb with it — a good live demonstration that the formula is real arithmetic and not a decoration.

06 Detection quality

This page says, in one paragraph: **this prototype measures how fast a crack is growing. It does not classify damage types, so there is no mAP to report** — and a number here that had not actually been measured would be worse than a blank.

Then it says what is measured instead: growth in millimetres per day, the quality of the fit behind it, and the lead time in days.

> **WHY AN EMPTY PAGE IS A STRONG PAGE**
>
> Half the prototypes in the room will quote a detection score they read in a paper. This one refuses to print a headline figure without a named test split and a stated sample size — the page is written so that it *cannot*. That is a claim about method, and it is the kind of claim a technical judge remembers.

When a classifier is trained, its score appears here per class, on a named split, with the sample size beside it. The published reference point to aim at is CRDDC'2022 — best team F1 0.769 on RDD2022. Quote that as *the target*, never as your result.

§ How it beats what already exists

	RAAHI	SURVEY VAN (NSV)	CRACK-DETECTION APP	COMPLAINT PORTAL
What it finds	**Cracks *and their growth rate***	Full distress survey	Cracks in one pass	Potholes, after failure
Question answered	**Which crack, and *when***	What condition is it in	Is there a crack here	Who complained loudest
Frequency	**Daily**	Periodic, often annual	Once per survey	Event-driven
Cost per km	**Near zero — the truck was already driving**	Very high	Low	Free, but biased
Repair verified	**Automatically, next pass**	None	None	Manual inspection
Coverage bias	**Fleet routes — reported openly**	Surveyed corridors only	Whoever drove it	Wealthier, vocal wards

Why has nobody built this?

Detecting a crack in one photograph is a *solved problem*. Measuring how fast that same crack is growing requires the same camera to return to the same metre of road every morning for weeks — which no survey vehicle, satellite or complaint app can do, and which a municipal waste fleet already does, for free, today.

THE LINE TO CLOSE ON

"A crack photographed once is a defect. Photographed for twenty mornings, it is a countdown."

§ The questions you will be asked

QUESTION	ANSWER
GPS is only accurate to 5–10 m. How is this the same crack?	It isn't, on GPS alone — and we say so on the page. Three signals vote: distance within 15 m, heading within 45°, and a 64-bit fingerprint of the scene itself. Position and heading together, or an unmistakable scene on its own. Every match prints its own reasoning.
How do we trust your millimetres?	You don't have to. Absolute length carries a constant bias; growth is a subtraction, so the bias cancels. Growth is the number we report. The scale itself comes from the lens by pure optics, and the app always names which source it used — including when it assumed one.
Where is the rainfall data from?	A bundled table of published monthly normals for Indian stations. The app labels it `normal`, never "forecast", because a normal says what a fortnight like this usually brings. Post a real IMD figure to the app and it uses that instead.
Your traffic numbers look made up.	They are class defaults, and every answer says `assumed` or `counted`. A municipality with real counts puts them in and the term changes in front of you.
What is your detection accuracy?	We have not measured one, so we do not quote one. This build measures growth. The page is written so it cannot print a score without a named split and a sample size.
What if it rains, or it is dark?	The camera pauses and resumes in daylight. Stated as a limitation on the overview tab rather than hidden.
Shadows and road markings?	Paint is present from day one and never grows. Anything that appears and spreads is damage. That single rule separates infrastructure from damage with no training data at all — and only a fleet that returns to the same metre every morning can apply it.
Two trucks photograph one crack.	Deduplicated by position, heading and visual match, so it is counted once however many vehicles pass it.
Nothing to compare on day one?	Correct. Detection works immediately; growth needs a baseline of roughly three weeks. The app shows grey, not green, until then — and says so.

Every figure on this page came from a real run of the prototype: eighteen photographs, one spot, 1.392 mm/day at R^2 0.974. Nothing was preloaded into the database.